Dataset Numbers:

Dataset	Describe the dataset — exact numbers	Why chosen / EDA / preprocessing
SkillCraft1	**3,395 examples (players)**; **8 classes** (LeagueIndex 1–8, Bronze → Professional); **20 columns total** in the raw dataset. After excluding the target, GameID, and the three incomplete variables (Age, HoursPerWeek, TotalHours), the current modelling set contains **15 numerical gameplay features**. Features include APM, ActionLatency, NumberOfPACs, hotkey use, map activity and unit-production behaviour.	Chosen as a **numerical player-behaviour dataset** for predicting skill. EDA shows clear relationships with skill: median **APM increases from 54.04 in Bronze to 274.34 in Professional**, while median **ActionLatency decreases from 93.33 to 35.41**. Correlations with LeagueIndex: **APM = +0.662**, **ActionLatency = −0.674**. Missing values occur in Age (**55**), HoursPerWeek (**56**) and TotalHours (**57**), so these were excluded from the initial clean feature set.
Dota2 Games Results	**102,944 examples (matches)**: **92,650 training + 10,294 test**. **2 target classes**: Team 1 win (1) or Team 2 win (−1). **117 columns total**, consisting of **1 target + 116 input features**: clusterid, gamemode, gametype, and **113 hero-selection features**. Hero features are discrete values: **−1, 0, +1**.	Chosen as a **large, sparse, discrete strategic-choice dataset** for match-outcome prediction. The training target is fairly balanced: **43,868 Team 2 wins vs 48,782 Team 1 wins**. There are **0 missing values**. Every match contains exactly **10 selected heroes: 5 per team**. The hero matrix is **91.15% sparse** because most of the 113 possible heroes are absent from each match. Visualisations: match outcome, game-mode/game-type distributions and top-15 most selected heroes.
Google Quick, Draw!	Our selected dataset contains **24,000 examples (drawings)** across **8 classes**: airplane, apple, bicycle, cat, fish, guitar, house and tree. Exactly **3,000 drawings per class**. Each image is **28 × 28 grayscale**, giving **784 pixel features** per drawing.	Chosen as the **visual/image component** of the project and for CNN experimentation. The original categories contain over **120,000 drawings each**, so we randomly sampled **3,000 per class using seed 42** to create a balanced dataset. Pixel values were normalized from **0–255 to 0–1** and reshaped from 784 values to **28×28×1**. We used a stratified **80/10/10 split**: **19,200 training, 2,400 validation, 2,400 test**, with every class represented equally. Visualisations: 8-class sketch grid and balanced class-distribution chart.